Exchange Scenario for the Origin of the Moon

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Imagine a binary with semimajor axis a and total mass M_b that contains the present Moon of mass M_m and a companion of mass M_c. Let $\eta = M_b/M_m > 1$ denote the total binary mass in units of the lunar mass. Because of the Sun's tidal gravity, a must less than the binary's effective Hill radius. At 1 AU from the Sun, this limiting radius is

$$r_{\mathrm{H},b} \equiv 1\,\mathrm{AU} \left(\frac{M_b}{3\,M_\odot}\right)^{1/3} \simeq 3.5 \times 10^5\, \eta^{1/3}\ \mathrm{km}\ . \tag{1}$$

The binary orbit can be no smaller than the present lunar radius of $\simeq$1700 km.

Suppose that the binary impinges on the Earth from some large distance with relative speed u and impact parameter b. We may treat the relative motion of the binary and Earth as a two-body problem only within the Earth's Hill sphere of radius $r_{\mathrm{H},e} \simeq 1.5 \times 10^6$ km, wherein the gravity of the Sun is a minor perturbation; other criteria, such as the radius of the "activity sphere," are quantitatively similar. The impact parameter and distance of closest approach s between the binary barycenter and the Earth are related by

$$b^2 = s^2(1 + 2\theta)\ . \tag{2}$$

Conservation of angular momentum provides a similar relation between u and the speed u_s at closest approach:

$$u_s^2 = u^2(1 + 2\theta) \tag{3}$$

Here θ is the Safronov parameter, given by

$$\theta = \frac{G(M_e + M_b)}{su^2} \simeq 4 s_5^{-1} u_0^{-2} \left(1 + \frac{M_b}{M_e}\right)\ , \tag{4}$$

where $M_e \simeq 6 \times 10^{27}$ g $\simeq 80\, M_m$ is the Earth's mass, $s_5 = s/10^5$ km, and $u_0 = u/1\,\mathrm{km\,s^{-1}}$. Hereafter, we will assume that $M_b/M_e \ll 1$.

If the binary is in a disk of planetesimals, then the characteristic relative speed is represented by the velocity dispersion σ of planetesimals in the coorbital frame of the Earth. The appropriate value for σ is unclear. If the heating of the planetesimal disk due to weak gravitational encounters is regulated by inelastic collisions, then we might expect σ to be of order the surface escape speed for the largest bodies in the system. A spherical object with uniform density ρ in $\mathrm{g\,cm^{-3}}$ and radius $R = 10^3\, R_3$ km has a surface escape speed of $\sim 1\, R_3 \rho^{1/2}\,\mathrm{km\,s^{-1}}$. Therefore, a plausible scale for the velocity dispersion is $\sigma \sim 1\,\mathrm{km\,s^{-1}}$ if the largest planetesimals are similar in size to the Moon.

In the regime $s < r_{\mathrm{H},e}$, we neglect the influence of the Sun's gravity and focus on how the Earth's tidal gravity modifies the dynamics of the binary. The tidal acceleration across the binary exceeds the binary self-gravity when the distance from the Earth is less than

$$s_t = a(M_e/M_b)^{1/3} \simeq 4.3\eta^{-1/3} a\ . \tag{5}$$

Tidal disruption of the binary occurs if $s \lesssim s_t$ and if sufficient time is spent near the minimum separation. The characteristic duration of the close approach is $\tau = s/u_s$. Clearly, τ goes to the hyperbolic value, b/u, as $\theta \to 0$. An approximate condition for disruption is that the magnitude of the relative velocity imparted to the binary components near s is $\gtrsim v = (GM_b/a)^{1/2} \simeq 0.22\,(\eta/a_5)^{1/2}\,\mathrm{km\,s^{-1}}$, the binary orbital speed. Mathematically, disruption is probable when

$$\frac{GM_e a}{s^3}\tau \gtrsim v\ . \tag{6}$$

Strongly focused incident binary trajectories ($\theta \gg 1$) have $\tau = (s^3/2GM_e)^{1/2}$, and it is easy to see that eq. (6) is satisfied for $s < s_t$, as we would expect. Energetic encounters with $\theta \ll 1$ spend less time within s_t and are less susceptible to disruption. In this limit, the largest incident speed that permits disruption is given by

$$u_{\rm dis} \sim v \left(\frac{s_t}{b}\right) \left(\frac{M_e}{M_b}\right)^{1/3} \simeq 4.3v \left(\frac{s_t}{b}\right) \eta^{-1/3} \ . \tag{7}$$

For $a_5 \lesssim 1$ and $s < s_t$, we have $u_{\rm dis} \gtrsim 1\,{\rm km\,s^{-1}}$, and thus tidal disruption is likely if $\sigma \sim 1\,{\rm km\,s^{-1}}$.

Disruption of the binary does not guarantee that one component is left bound to the Earth. An approximate condition for capture can be formulated as follows. Suppose that the binary is disrupted and that, in the rest frame of the Earth, M_m receives a velocity kick of magnitude $(M_c/M_b)v$ in a direction opposite to the motion of the binary at closest approach. The new specific energy of the Moon relative to the Earth is then

$$E_m \simeq \frac{1}{2}\left(u_s - q_c v\right)^2 - \frac{GM_e}{s} \simeq \frac{1}{2}u^2\left(1 - 2q_c\frac{u_s v}{u^2}\right) \ , \tag{8}$$

where $q_c = M_c/M_b = (1-\eta)/\eta$, $u^2/2 = u_s^2/2 - GM_e/s$, and we neglected the v^2 term for simplicity. Capture occurs if $E_m < 0$, which is equivalent to $2q_c u_s v/u^2 > 1$. The maximum u that permits capture for given a and s is

$$u_{\rm cap}^2 = 2q_c^2 v^2 \left[1 + \left(1 + \frac{2GM_e}{s q_c^2 v^2}\right)^{1/2}\right] \ . \tag{9}$$

When $\eta \sim 1$ the second term under the square root is $\gg$1 and we find

$$u_{\rm cap} \simeq 2^{3/4} q_c^{1/2} \left(\frac{M_e a}{M_b s}\right)^{1/4} v \simeq 5\eta^{-2/3}(\eta - 1)^{1/2}\left(\frac{s_t}{s}\right)^{1/4} v \ . \tag{10}$$

This is precisely the result obtained the in the limit of $\theta \gg 1$. In general, we expect that $u_{\rm cap}$ is a few times v. The limiting u for capture appears to be more restrictive than the disruption condition in eq. (7). Nonetheless, it seems that $u_{\rm dis}/\sigma$ may not be too different from unity, in which case both disruption and capture may be rather probable.

If the Moon is captured by this mechanism, it is straightforward to crudely estimate the initial parameters of its orbit about the Earth. The semimajor axis is $a_m = -GM_e/2E_m$. If we let $E_m = -\epsilon u^2$, then

$$a_m = \frac{GM_e}{2\epsilon u^2} = 2 \times 10^5 \epsilon^{-1} \left(\frac{u}{1\,{\rm km\,s^{-1}}}\right)^{-2} \ . \tag{11}$$

The orbit of the newly captured Moon is susceptible to instability and disruption if $a_m \gtrsim r_{\rm H,e} \sim 10^6$ km. Given a_m, the eccentricity e_m may be estimated by assuming that the pericenter distance equals s, so that

$$1 - e_m \simeq \frac{s}{a_1} \simeq 0.5\, s_5\, \epsilon \left(\frac{u}{1\,{\rm km\,s^{-1}}}\right)^2 \ . \tag{12}$$